DATA HÔPITAL

1/ Compréhension des métiers:

- Objectif Métier (Détecter les caractéristiques):
  - **Améliorer la qualité des soins médicaux fournis par l'hôpital.**
  - Optimiser les processus internes
  - Personnaliser les traitements des patients
  - Réduire les risques de maladies cardiovasculaires
- Objectif Data Mining:

L'objectif du data mining est d'extraire à partir des données des patients des relations entre les données et créer des clusters et classes qui nous aideront dans la classification des patients.

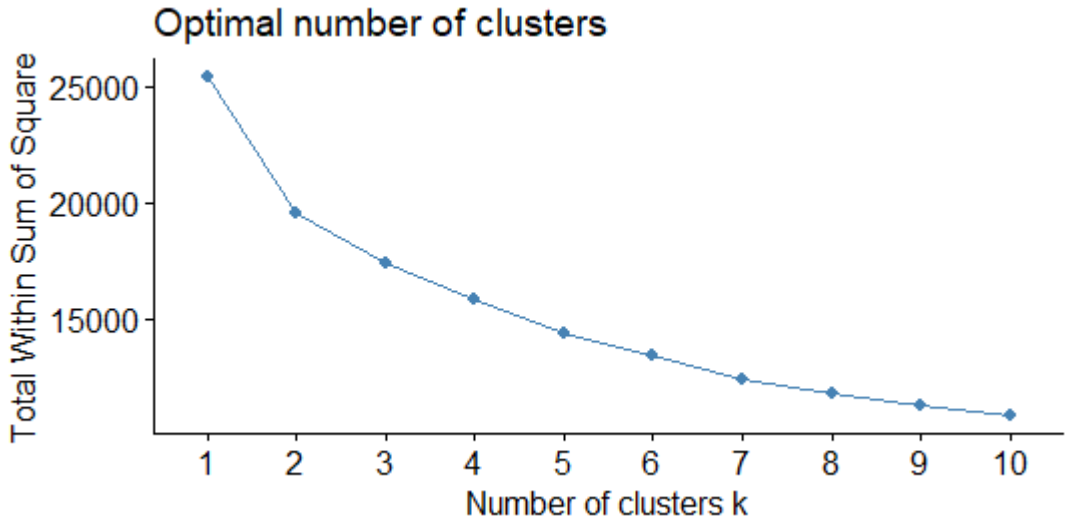
2/Compréhension des données:

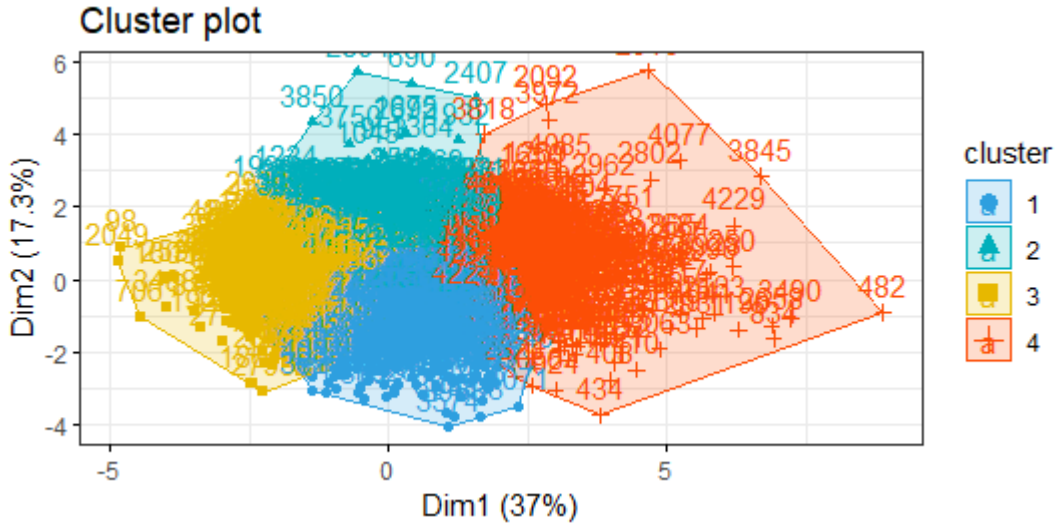
- **Nettoyage des données :** Identifier et traiter les valeurs manquantes, les valeurs aberrantes …
- **Préparation des données :** Normalisation, encodage des variables …

3/Préparation des données:

```
> summary(hopital)
      male             age          education     currentSmoker      cigsPerDay         BPMeds       
 Min.   :0.0000   Min.   :32.00   Min.   :1.000   Min.   :0.0000   Min.   : 0.000   Min.   :0.00000  
 1st Qu.:0.0000   1st Qu.:42.00   1st Qu.:1.000   1st Qu.:0.0000   1st Qu.: 0.000   1st Qu.:0.00000  
 Median :0.0000   Median :49.00   Median :2.000   Median :0.0000   Median : 0.000   Median :0.00000  
 Mean   :0.4292   Mean   :49.58   Mean   :1.979   Mean   :0.4941   Mean   : 9.006   Mean   :0.02962  
 3rd Qu.:1.0000   3rd Qu.:56.00   3rd Qu.:3.000   3rd Qu.:1.0000   3rd Qu.:20.000   3rd Qu.:0.00000  
 Max.   :1.0000   Max.   :70.00   Max.   :4.000   Max.   :1.0000   Max.   :70.000   Max.   :1.00000  
                                  NA's   :105                      NA's   :29       NA's   :53       
 prevalentStroke     prevalentHyp       diabetes          totChol          sysBP           diaBP       
 Min.   :0.000000   Min.   :0.0000   Min.   :0.00000   Min.   :107.0   Min.   : 83.5   Min.   : 48.0  
 1st Qu.:0.000000   1st Qu.:0.0000   1st Qu.:0.00000   1st Qu.:206.0   1st Qu.:117.0   1st Qu.: 75.0  
 Median :0.000000   Median :0.0000   Median :0.00000   Median :234.0   Median :128.0   Median : 82.0  
 Mean   :0.005896   Mean   :0.3106   Mean   :0.02571   Mean   :236.7   Mean   :132.4   Mean   : 82.9  
 3rd Qu.:0.000000   3rd Qu.:1.0000   3rd Qu.:0.00000   3rd Qu.:263.0   3rd Qu.:144.0   3rd Qu.: 90.0  
 Max.   :1.000000   Max.   :1.0000   Max.   :1.00000   Max.   :696.0   Max.   :295.0   Max.   :142.5  
                                                      NA's   :50                                     
      BMI          heartRate         glucose         TenYearCHD      
 Min.   :15.54   Min.   : 44.00   Min.   : 40.00   Min.   :0.0000  
 1st Qu.:23.07   1st Qu.: 68.00   1st Qu.: 71.00   1st Qu.:0.0000  
 Median :25.40   Median : 75.00   Median : 78.00   Median :0.0000  
 Mean   :25.80   Mean   : 75.88   Mean   : 81.96   Mean   :0.1519  
 3rd Qu.:28.04   3rd Qu.: 83.00   3rd Qu.: 87.00   3rd Qu.:0.0000  
 Max.   :56.80   Max.   :143.00   Max.   :394.00   Max.   :1.0000  
 NA's   :19      NA's   :1        NA's   :388                      
> 
```

- Remplacer les valeurs manquantes par NULL
- Convertir les variables binaires en facteurs





4/Modélisation:

Histogrammes pour les distributions (Age, sysbp, diabp,bmi)

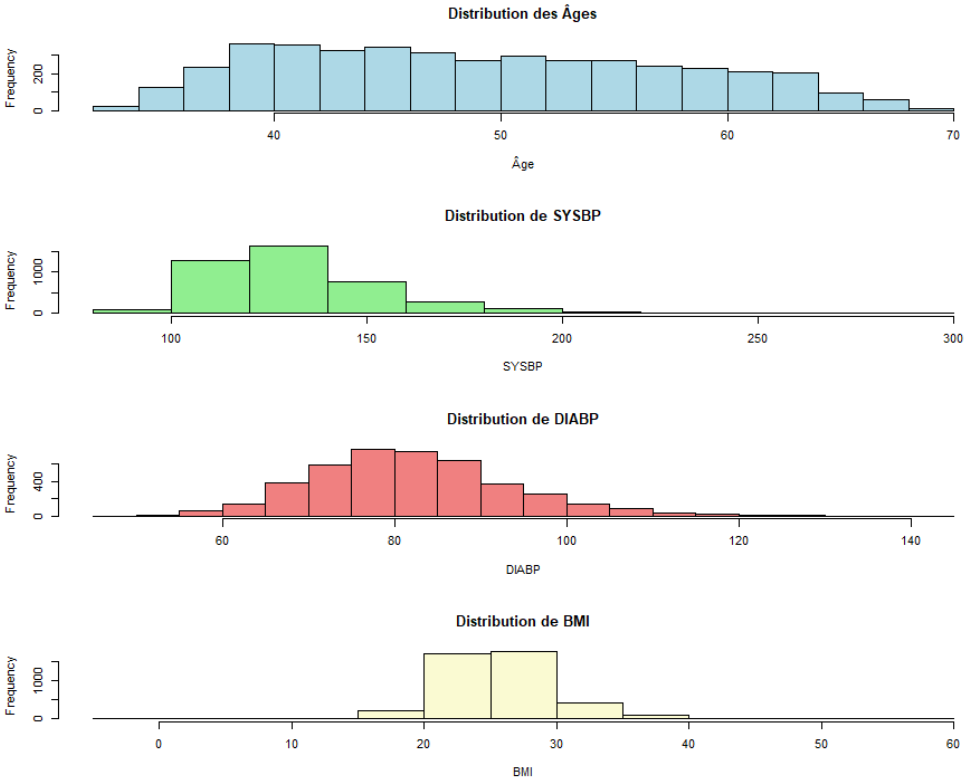
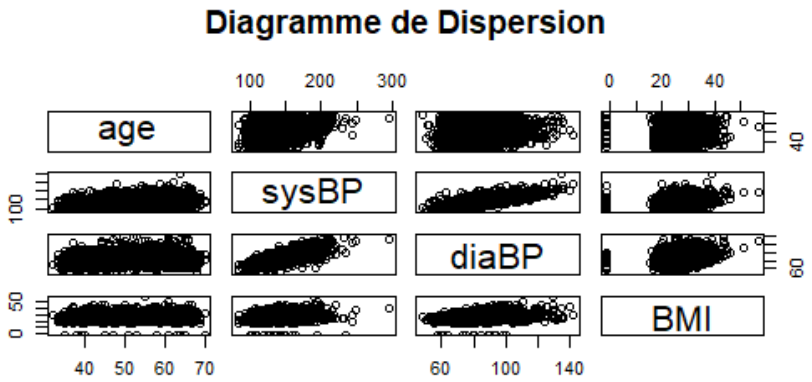
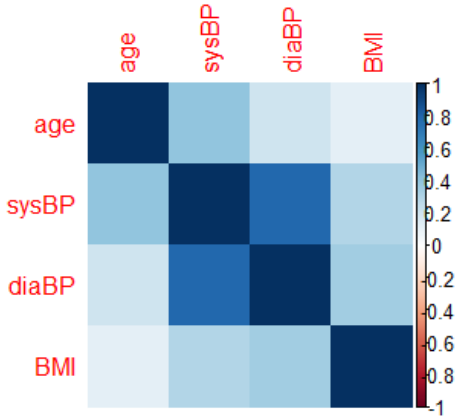


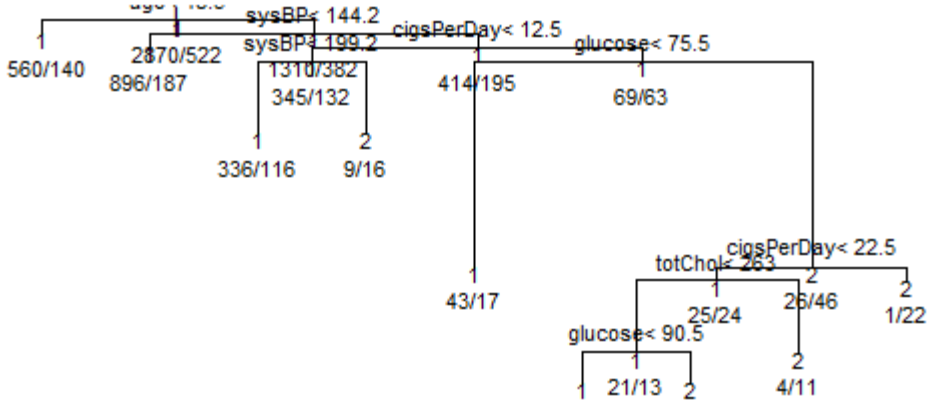
Diagramme de dispersion



Matrice de corrélation



Arbre de décision:




```
> conf_matrix_tree <- confusionMatrix(pred_tree,
> conf_matrix_tree
Confusion Matrix and Statistics

          Reference
Prediction   0   1
         0 691 123
         1  25   9

               Accuracy : 0.8255
                 95% CI : (0.7982, 0.8504)
    No Information Rate : 0.8443
    P-Value [Acc > NIR] : 0.9392

                  Kappa : 0.0477

 Mcnemar's Test P-Value : 1.544e-15

            Sensitivity : 0.96508
            Specificity : 0.06818
         Pos Pred Value : 0.84889
         Neg Pred Value : 0.26471
             Prevalence : 0.84434
         Detection Rate : 0.81486
   Detection Prevalence : 0.95991
      Balanced Accuracy : 0.51663

       'Positive' Class : 0

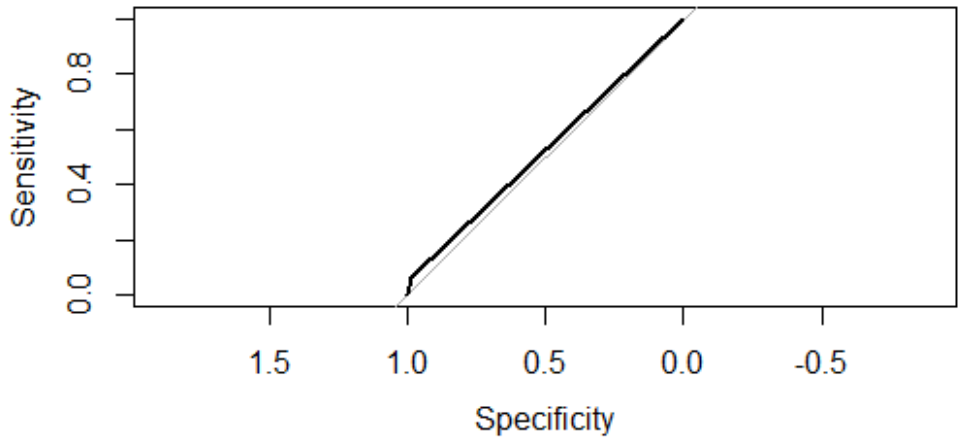
> |
```

5/Evaluation: tester l'algorithme

```
> conf_matrix_tree

pred_tree   1   2
        1 714 114
        2  12   8
> |
```

Courbe ROC




```
> auc(roc_curve)
Area under the curve: 0.5245
>
```

```
> summary(modele_lineaire)

Call:
lm(formula = TenYearCHD ~ age + education + currentSmoker + cigsPerDay +
    BPMeds + prevalentStroke + prevalentHyp + diabetes + totChol +
    sysBP + diaBP + BMI + heartRate + glucose, data = train_data)

Residuals:
     Min       1Q   Median       3Q      Max
-0.63169 -0.18429 -0.10226 -0.02172  1.05740

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)       4.845e-01  8.757e-02   5.533 3.43e-08 ***
age               7.425e-03  8.617e-04   8.617  < 2e-16 ***
education        -9.402e-04  6.372e-03  -0.148  0.88270
currentSmoker     6.880e-03  2.026e-02   0.340  0.73425
cigsPerDay        2.637e-03  8.524e-04   3.094  0.00199 **
BPMeds1           5.352e-02  3.940e-02   1.358  0.17444
prevalentStroke1  1.839e-01  8.670e-02   2.121  0.03400 *
prevalentHyp1     3.742e-02  1.949e-02   1.920  0.05497 .
diabetes1         8.034e-03  4.946e-02   0.162  0.87097
totChol           9.012e-05  1.507e-04   0.598  0.54981
sysBP             2.198e-03  5.497e-04   3.999 6.51e-05 ***
diaBP            -1.098e-03  9.021e-04  -1.217  0.22358
BMI               1.742e-03  1.733e-03   1.006  0.31466
heartRate        -1.021e-03  5.461e-04  -1.870  0.06158 .
glucose           8.529e-04  3.444e-04   2.477  0.01331 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.3412 on 2910 degrees of freedom
Multiple R-squared:  0.09197,   Adjusted R-squared:  0.0876
F-statistic: 21.05 on 14 and 2910 DF,  p-value: < 2.2e-16
```